PT4052 | PT7004 | ST7050

TENDON INJURIES

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Management of tendon injuries

Discuss	Discuss the incidence, mechanisms of tendon injuries
Describe	Describe Tendon Injury Healing
Discuss	Discuss evidence-based management approaches for tendon injuries and debate pros and cons for each
Design	Design a management plan for case studies

MUSCULOSKELETAL ANATOMY TIPS

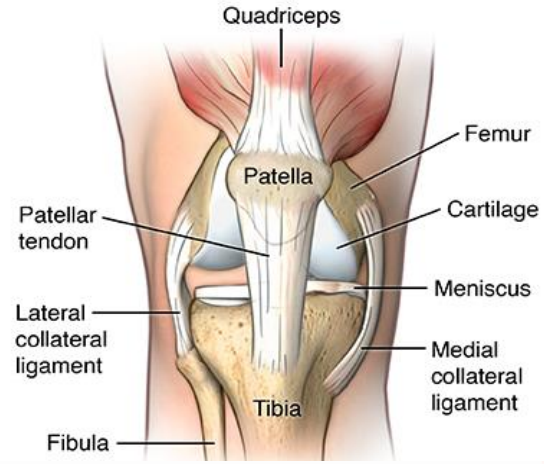


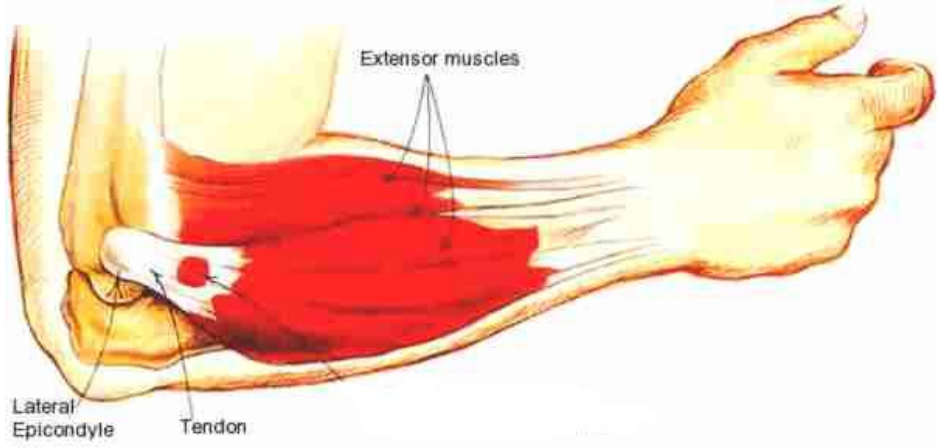


Function

- Connect muscle to bone to allow movement of a joint
- Pass forces from muscle to bone
- Non-contractile tissue
- Store energy that is released to increased output
- Provide sensory proprioceptive feedback



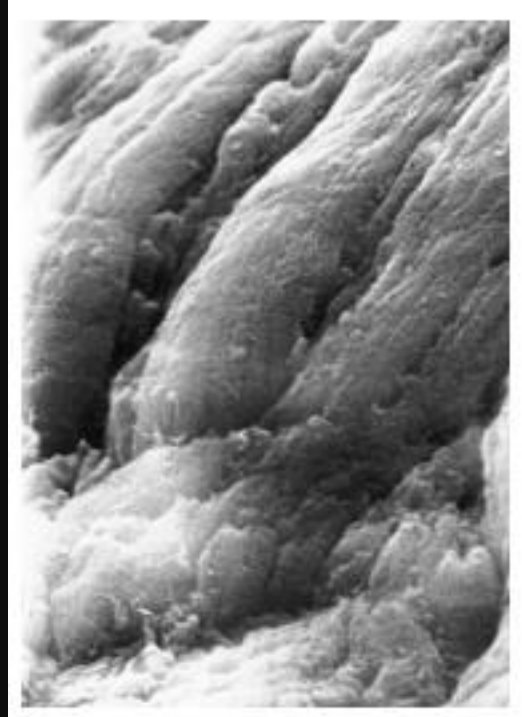


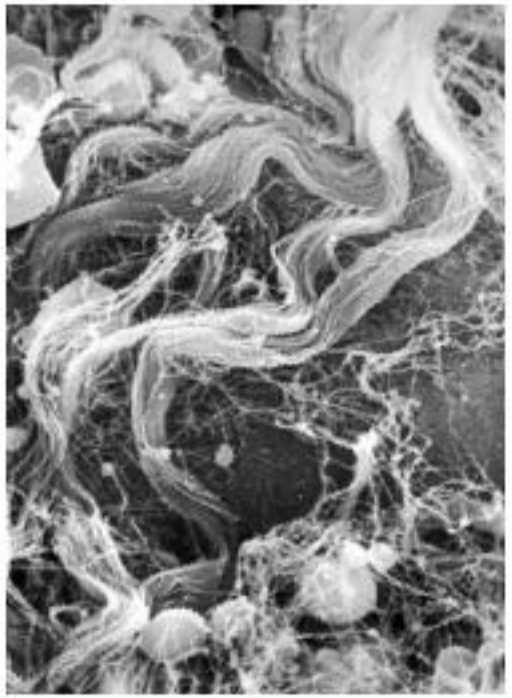


https://www.youtube.com/watch?v=K9rQ9aDtzrk&t=10s

Factors affecting properties of tendon

- Aging
  - Size
  - Water content
  - Collagen quality
- Pregnancy
- Movement
- Immobilisation
- Diabetes
- Steroids





Tendon Injuries Cost

Bergamin, F., Civera, M., Reinoso, M. R., Burgio, V., Ruiz, O. G., & Surace, C. (2023). Worldwide Incidence and Surgical Costs of Tendon Injuries: A Systematic Review and Meta-Analysis. Muscles, Ligaments & Tendons Journal (MLTJ), 13(1).

Table V. Cost of tendon repair.

	Surgical cost (€)	Total tendon ruptures	Surgical treatments	Total surgical cost (€)	Total surgical cost per year (€) (rounded)
Surveyed countries					
Sweden	7,332 (38)	12,000	7,200	52,790,400	52.80 mil
Finland	1,259 (39)	5,000	3,000	3,777 000	3.78 mil
Turkey	1,772 (40)	66,000*	39,600	70,171,200	70.17 mil
State of New York	12,764 (41)	18,000	10,800	137,851,200	137.85 mil
Canada	16,705 (42)	38,000	22,800	380,874,000	380.87 mil
New Zealand	822 (29)	4,000	2,400	1,972,800	1.97 mil
World zones					
Europe	3,454	411,000	246,600	851,756,400	8.52 mil
North America	14,734	347,000	208,200	3,067,618,800	3.07 bil
Oceania	822	18,000	10,800	8,877,600	8.88 mil
World					
I	6,337	6,697,000	4,018,200	25,463,333,400	25.46 bil
II	6,776	7,008,000	4,204,800	28,491,724,800	28.49 bil

Reported data: elaboration and results for the reported countries, the world zones and the world.

Achilles Tendon Rupture https://www.youtube.com/watch?v=CdbJmKB0buk

Achilles Tendon Rupture

https://www.youtube.com/watch?v=QUJgT-loxgc



Acute tendon injuries

- Signs & symptoms:
- Partial rupture or tear:
  - Sudden onset of pain
  - Local tenderness
  - Swelling

- Total rupture:
- Sudden onset of symptoms, no pain
- Loud snap
- Requires surgery and progressive rehabilitation



Avulsion Injuries

Acute avulsion injury results from extreme, unbalanced and often eccentric muscular contractions, and patients with such injuries present with severe pain and loss of function.

Adolescents are particularly vulnerable to avulsion injuries because of the inherent weakness of the apophyses. The many apophyses in the pelvis and hip are common sites of avulsion injuries. The single most common site of apophyseal avulsion is at the ischial tuberosity.

Cheerleaders, sprinters, gymnasts, track athletes, American football players, and baseball players are commonly affected.

Treatment for avulsion injury is generally conservative and the prognosis is good, but non-union may occur.

Tendinopathy

- Tendinopathy is an umbrella term used to describe tendon pain without knowing the specific pathology.
- Tendinitis implies an inflammatory response.
- However injured tendons appear to be in a degenerative state with few inflammatory cells.
- Tendinosis is a better term for the degenerative state of an injured tendon.

Charnoff J, Ponnarasu S, Naqvi U. Tendinosis. [Updated 2022 Oct 17]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2024 Jan-. Available from: https://www.ncbi.nlm.nih.gov/books/NBK448174/

Tendinopathy Prevalence

- Achilles tendon pathology is common in active people [Maffulli, 2019]. Achilles tendinopathy: Has an increasing incidence, which may be due to increased sporting activity and longer and more intense training [Ackermann, 2018a].
- It is estimated to affect more than 150,000 people in the UK every year [Kearney, 2013].
- Is more common in runners, especially middle-aged, middle-distance, and long-distance runners [Ackermann, 2018a].

https://cks.nice.org.uk/topics/achilles-tendinopathy/background-information/prevalence/#:~:text=It%20is%20estimated%20to%20affect,years%20%5BAlbers%2C%202016%5D.

Tendinopathy Prevalence

- Occurs in sports other than running. The estimated prevalence in dancers is 9%, in gymnasts 5%, in tennis players 2%, and in football players less than 1% [Wilson, 2010].
- Insertional Achilles tendinopathy is more common in older people, retired athletes, or less athletic, overweight people [Maffulli, 2019].
- Achilles tendon rupture:
  - Is common — around 4500 people in the UK seek medical help for this injury each year [Boyd, 2015].
  - Has an annual incidence of approximately 18 per 100,000 people [Uquillas, 2015].
  - Has a mean age of 37–43.5 years [Uquillas, 2015].

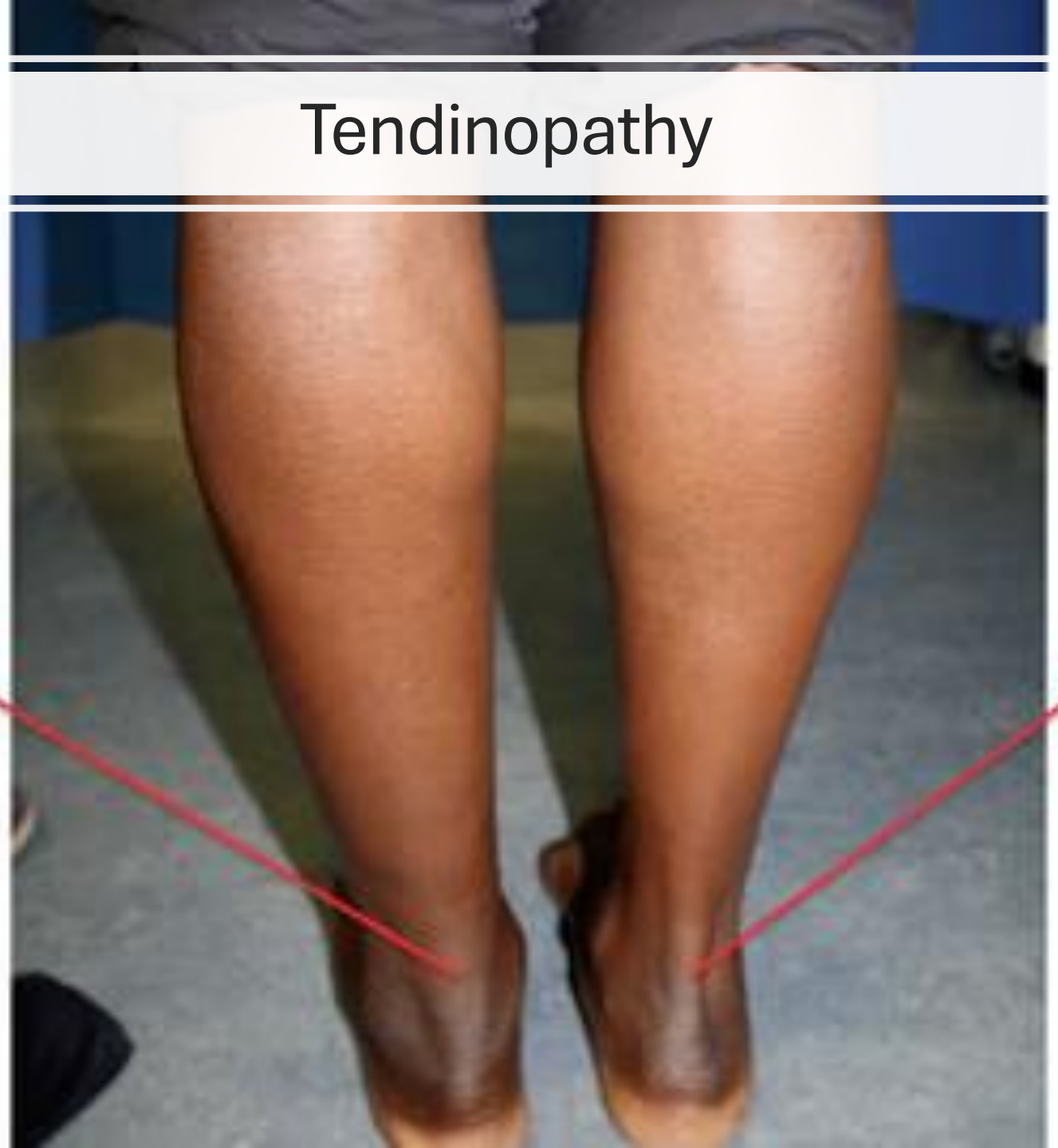
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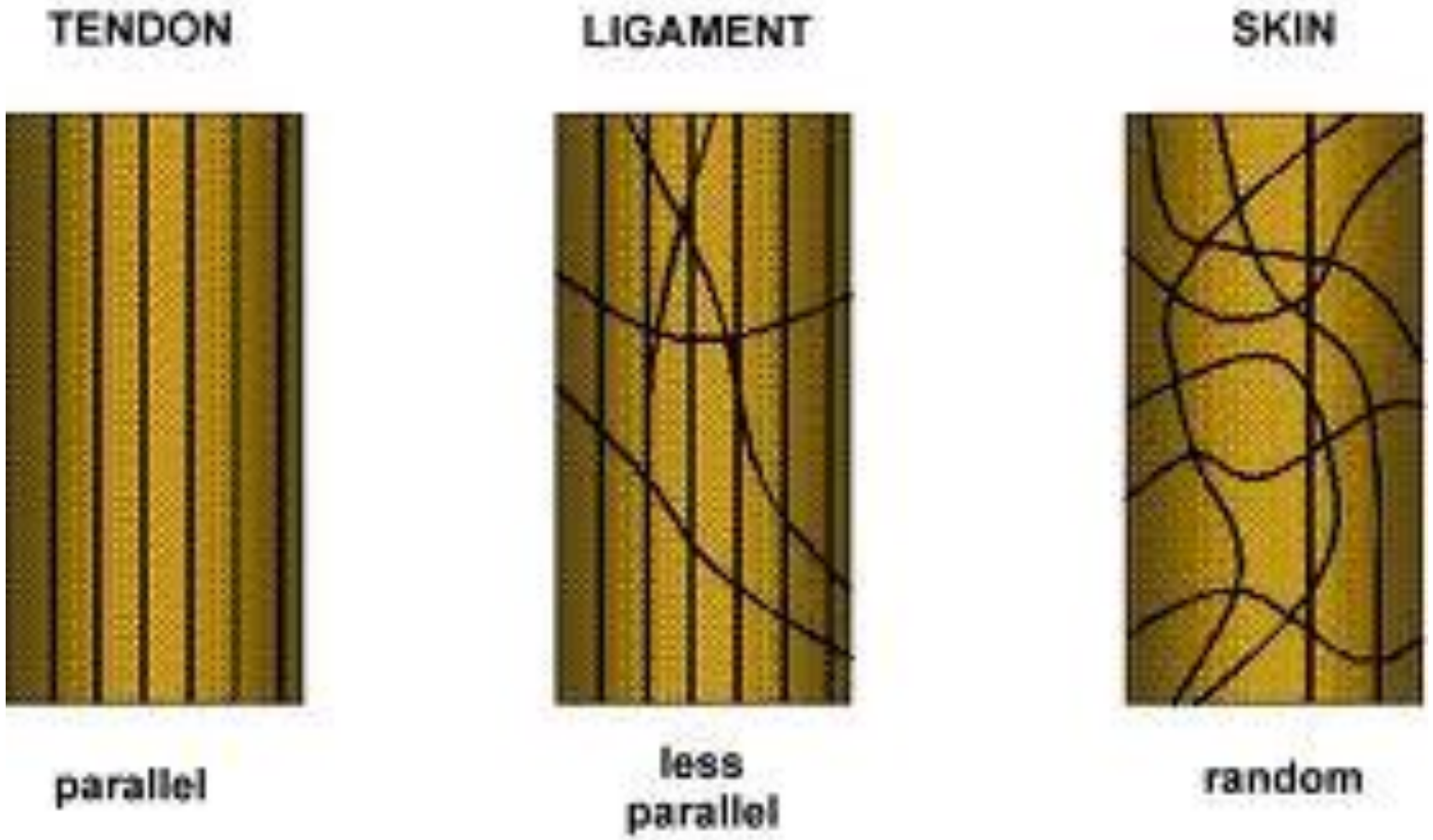
Tendinopathy

https://e3rehab.com/achilles-tendinopathy/

Tendinopathy



Tendon vs Ligament

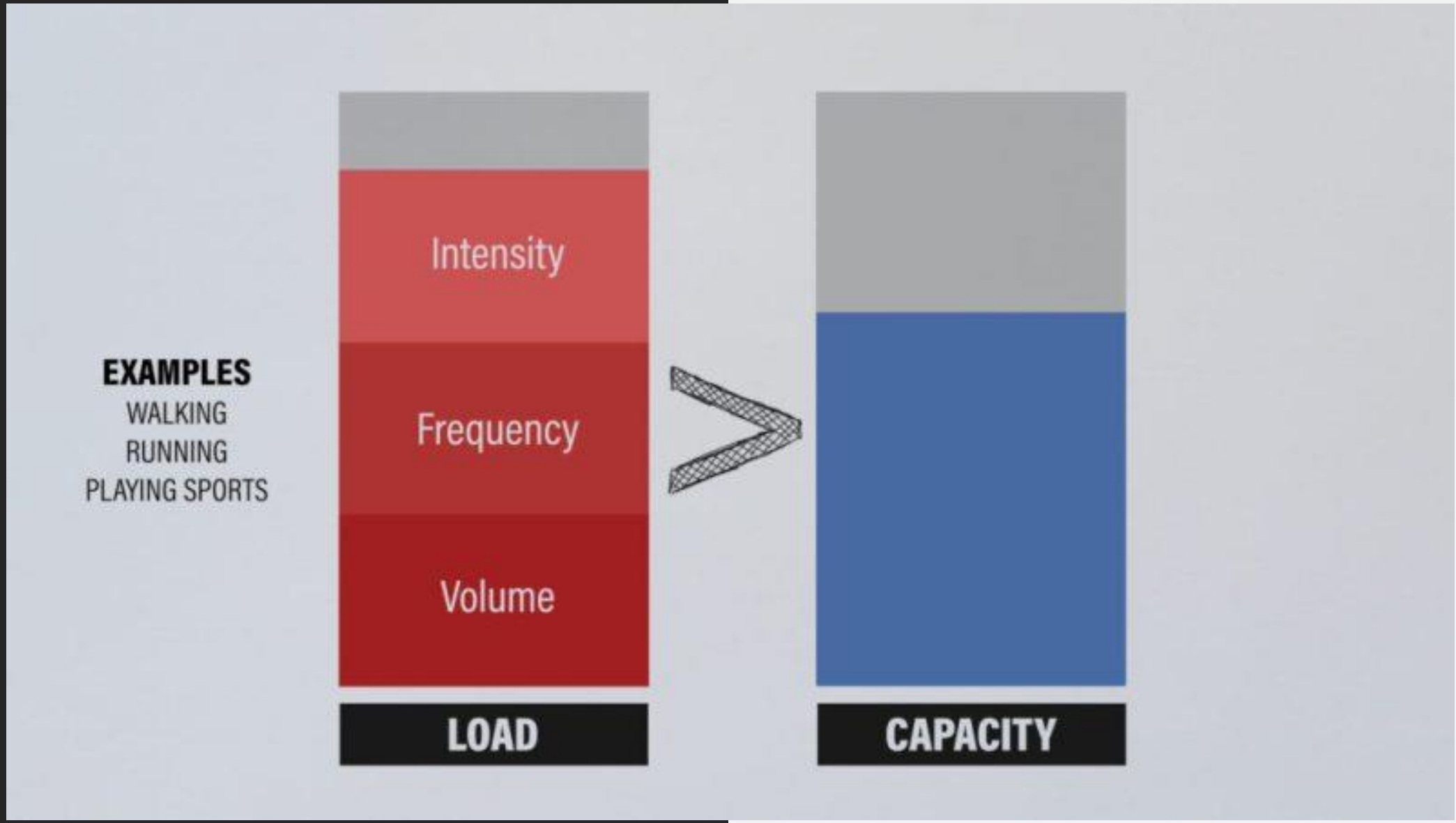


Tendon vs Ligament

Component	Ligament	Tendon
SOLIDS		
Collagen	**70-80%**	**Slightly higher**
- Type I	**90%**	**95-99%**
- Type III	**10%**	**1-5%**
- Ground substance (Proteoglycans, glycoproteins, plasma proteins)	**20-30%**	**Slightly less**
- Elastin (small amounts)		

Tendon injuries

- Load exceeds physiological range –> micro failure
- If load continues – gross failure
- Tendon stress is from muscle contraction
  - Maximal contraction with rapid eccentric load exceeds norm (and no reflex relaxation)
- Tendon strength is around twice that of the attaching muscle – hence muscle ruptures more common



Tendinopathy

https://e3rehab.com/achilles-tendinopathy/

Vascularity

- Limited vascularity – 7.5 x less than skeletal muscle
  - Therefore reduced healing properties as lower metabolic rate

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- Vascular supply from;
  1. Myotendinous junction
  2. Osteotendinous junction
  3. Paratendon/Synovial sheath

Intrinsic

Extrinsic

Vascularity

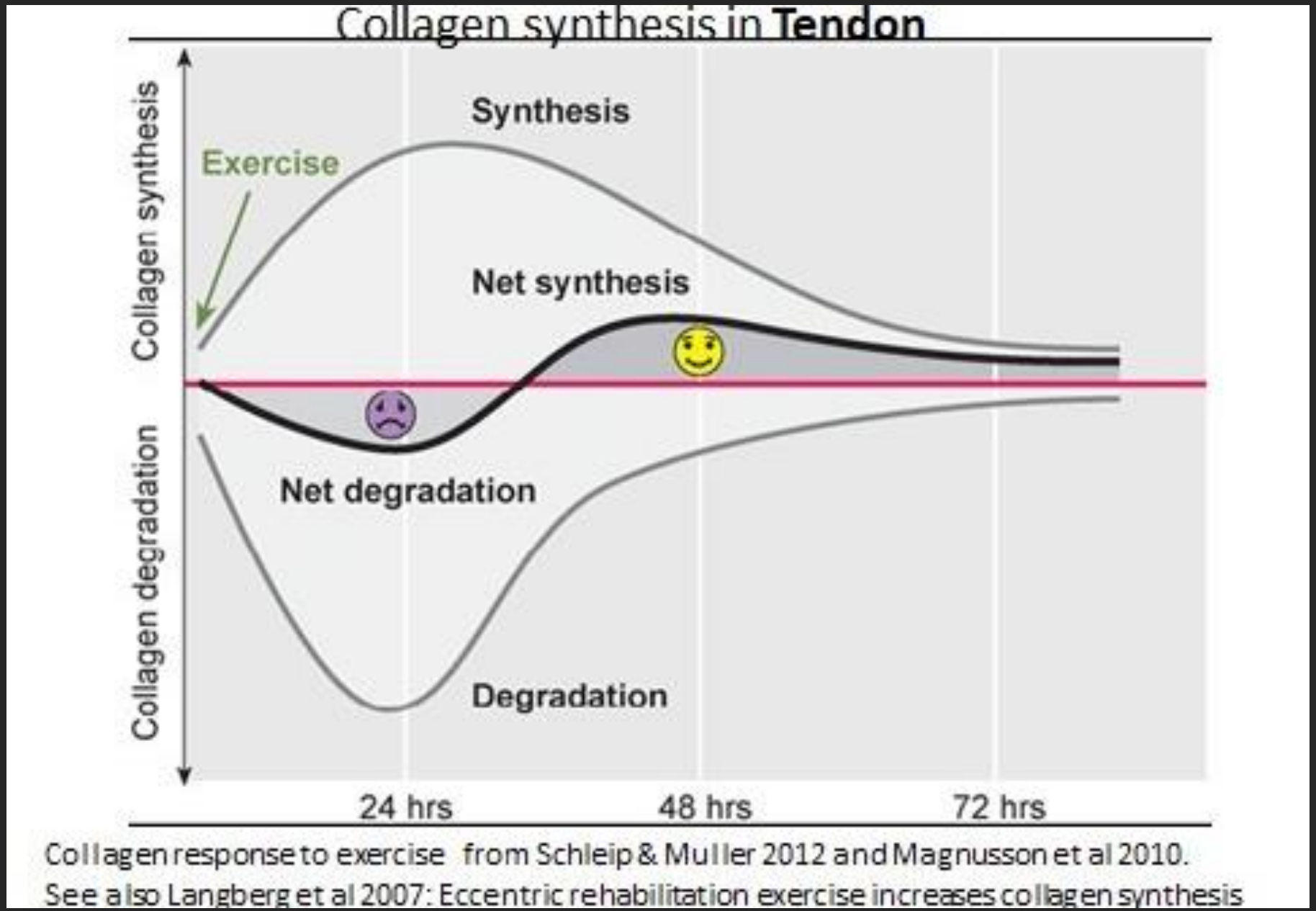
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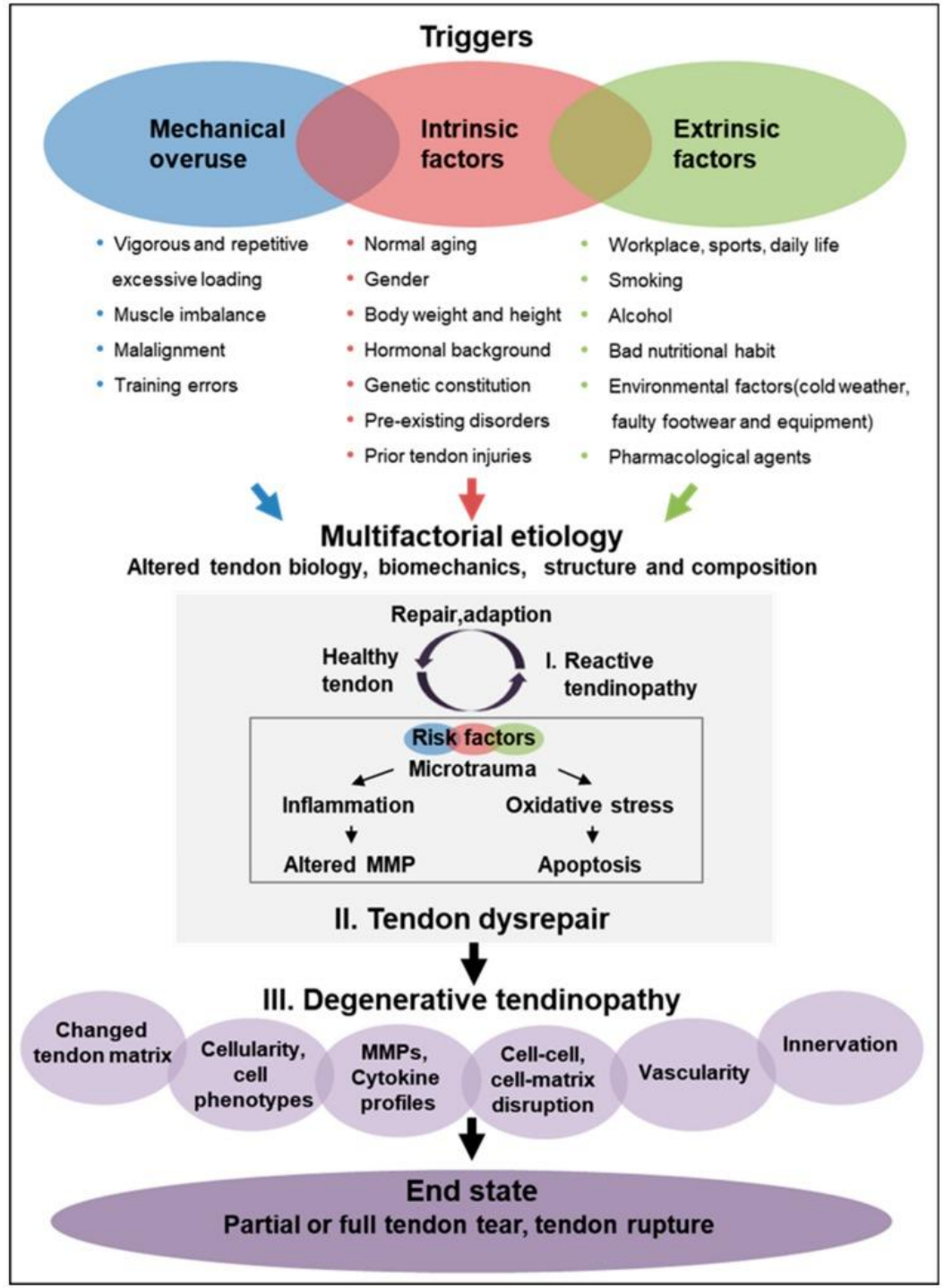
  Intrinsic

  Extrinsic

- Tendons also receive nutrition from synovial diffusion



Collagen response to exercise from Schleip & Muller 2012 and Magnusson et al 2010.
See also Langberg et al 2007: Eccentric rehabilitation exercise increases collagen synthesis


Triggers
Mechanical overuse
Intrinsic factors
Extrinsic factors
• Vigorous and repetitive excessive loading
• Muscle imbalance
• Malalignment
• Training errors
• Normal aging
• Gender
• Body weight and height
• Hormonal background
• Genetic constitution
• Pre-existing disorders
• Prior tendon injuries
• Workplace, sports, daily life
• Smoking
• Alcohol
• Bad nutritional habit
• Environmental factors(cold weather, faulty footwear and equipment)
• Pharmacological agents
Multifactorial etiology
Altered tendon biology, biomechanics, structure and composition
Repair,adaption
Healthy tendon
I. Reactive tendinopathy
Risk factors
Microtrauma
Inflammation
Oxidative stress
Altered MMP
Apoptosis
II. Tendon dysrepair
III. Degenerative tendinopathy
Changed tendon matrix
Cellularity, cell phenotypes
MMPs, Cytokine profiles
Cell-cell, cell-matrix disruption
Vascularity
Innervation
End state
Partial or full tendon tear, tendon rupture

Triggers



- Vigorous and repetitive excessive loading
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- Malalignment
- Training errors

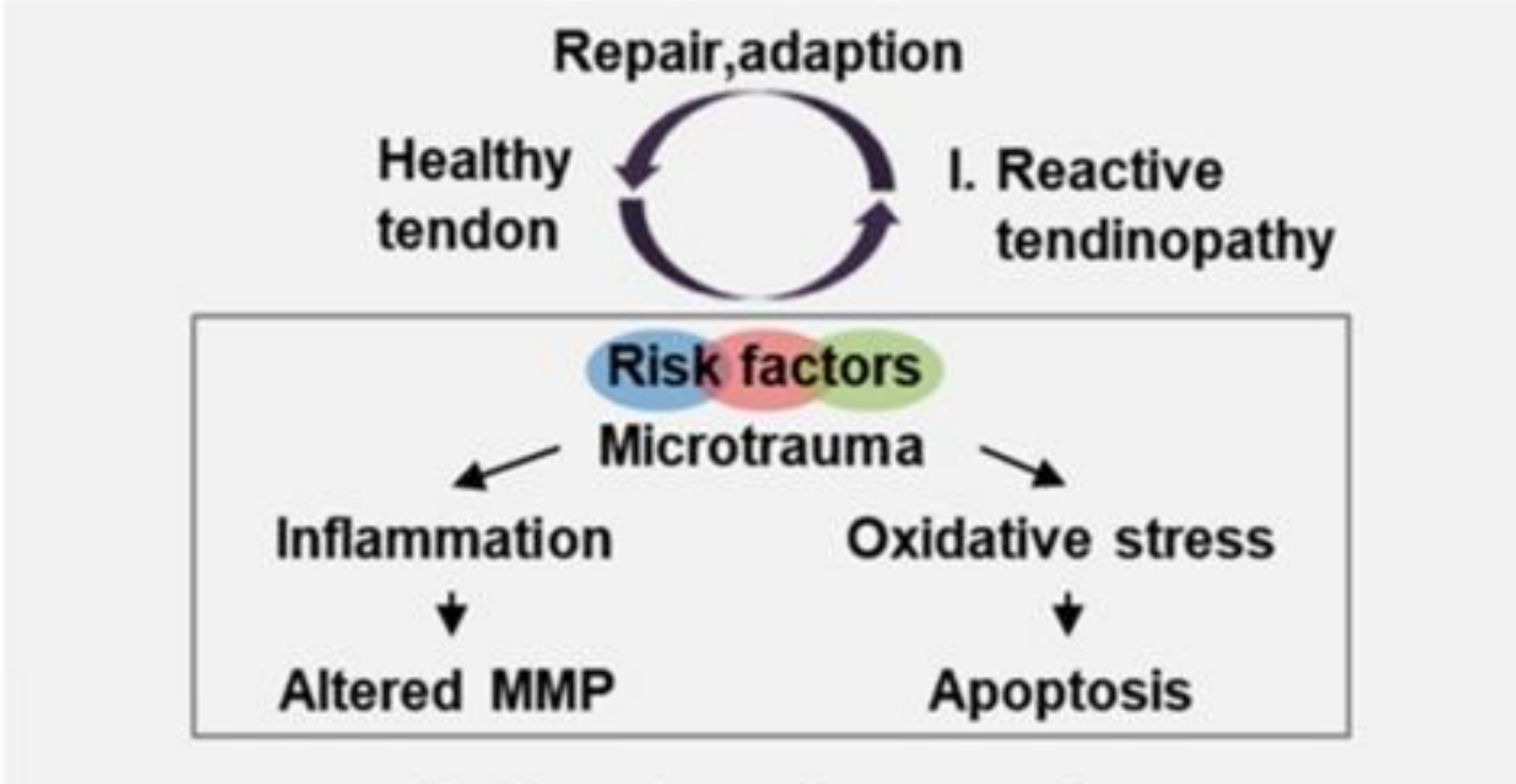
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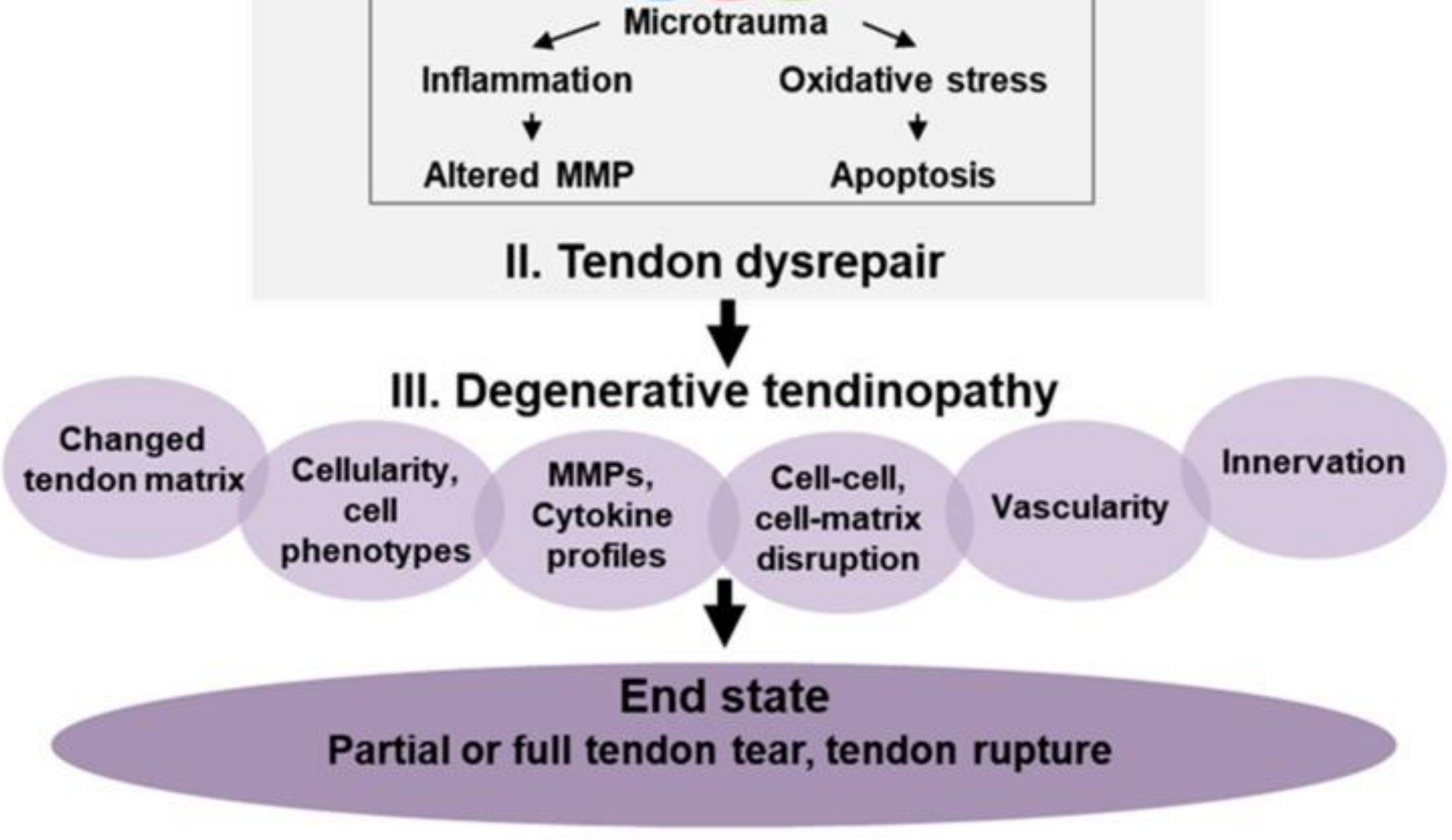
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- Genetic constitution
- Pre-existing disorders
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Multifactorial etiology

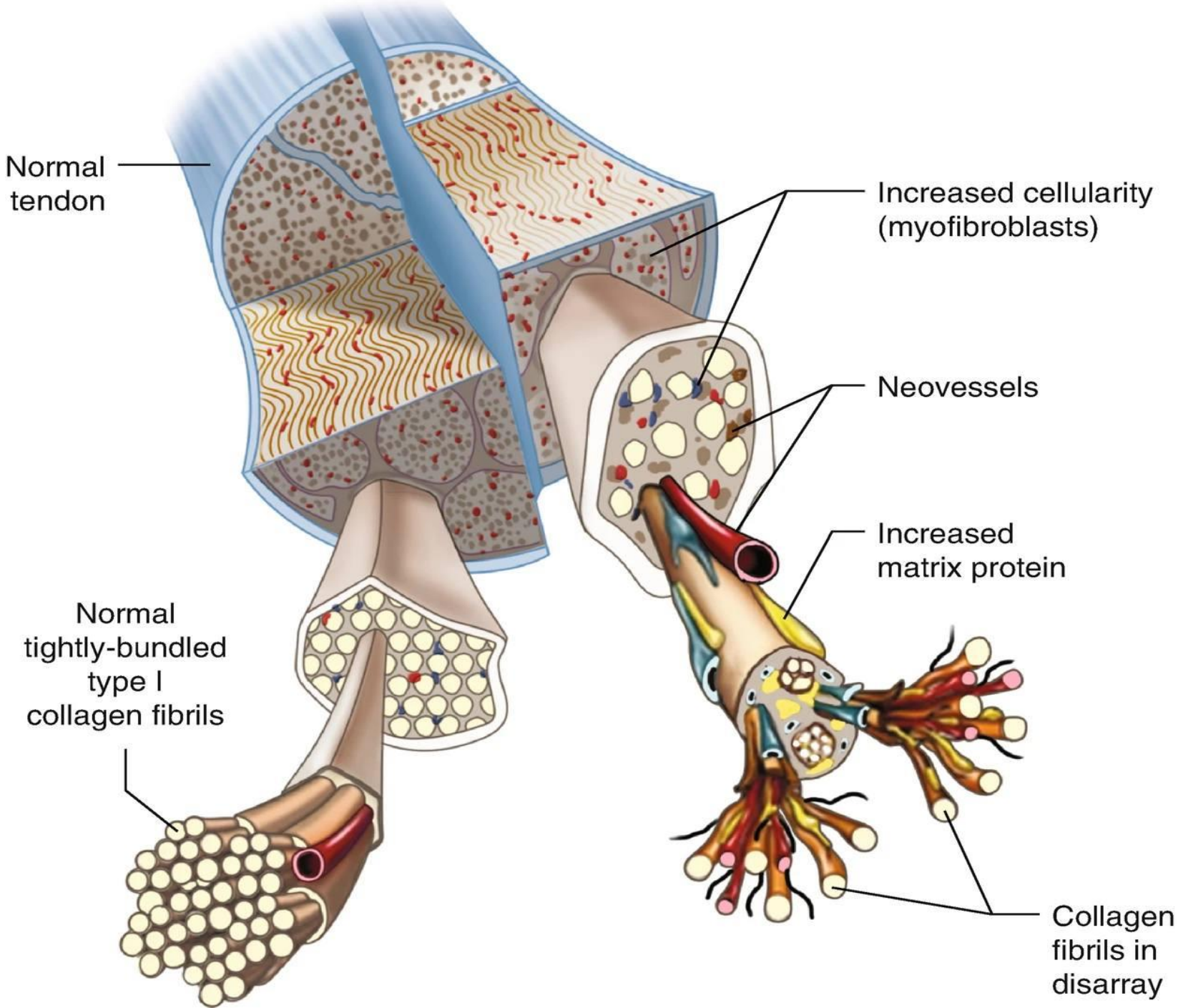
Altered tendon biology, biomechanics, structure and composition




Microtrauma
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Oxidative stress
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Changed tendon matrix
Cellularity, cell phenotypes
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Cell-cell, cell-matrix disruption
Vascularity
Innervation
End state
Partial or full tendon tear, tendon rupture

Tendinosis

- Signs & symptoms:
  - Pain after exercise, or following morning
  - Can be painful at rest
  - Initially becomes less painful with use
  - Pain can disappear when warming up, pain returns after exercise
  - Local tenderness and thickening
  - Possibly swelling and crepitus
- Common sites:
  - Rotator cuff
  - Extensor carpi radialis brevis
  - Patellar tendon
  - Achilles tendon



Summary based on Scott (2015) https://pubmed.ncbi.nlm.nih.gov/26390273/

Patology of tendons

Increased thickness

In-growth of vessels

In-growth of nerve endings

Pain

Reduced function

Diagnosis

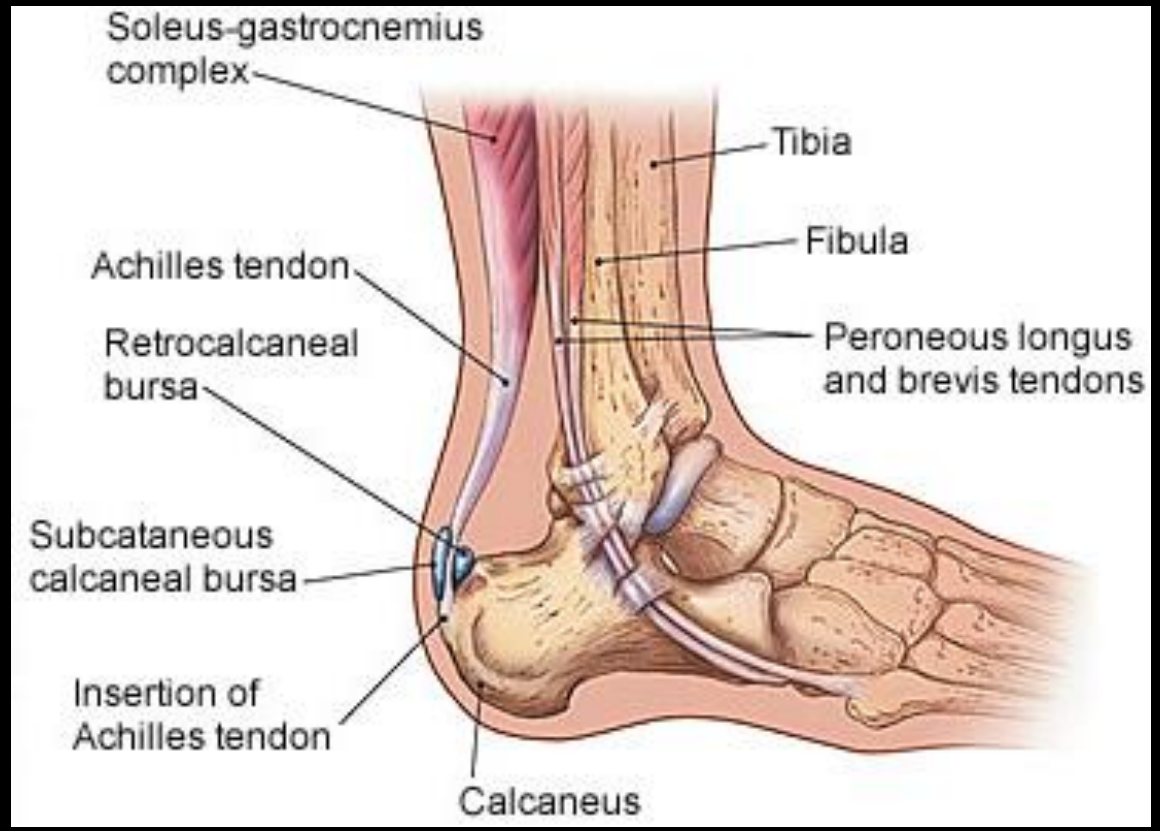
- Tendinosis can be difficult to differentiate from partial tears
- History is a key factor
  - Pain at rest – becomes less painful with use, run through pain, returns after, tenderness and thickening, crepitus
  - Sudden pain during explosive activity, had to slow, palpable inflam location
- Acute or chronic onset?
- Location may aid diagnosis (e.g. tenosynovitis usually near bone prominance)
- Palpation and passive movement
- MRI/CT/USS can be useful but also limited

Tenosynovitis / Tenovaginitis

- Tenosynovitis is a broad term describing the inflammation of the fluid-filled synovium within the tendon sheath. Can be have overuse, autoimmune, idiopathic, and infective causes.
  - Difficulty moving the affected joint.
  - Tenderness of the tendon sheath
  - Enlargement of the affected tendon in a symmetric pattern
  - Baseline contracture of the affected tendon
  - Pain with passive straightening of the tendon

Paratendonitis

- Irritation of tendon paratenon (Paratenonitis): Inflammation of the tendon/sheath that covers the tendon
- Occurs when paratenon rubs over bony prominence
- Occurs with partial tears and tendinosis



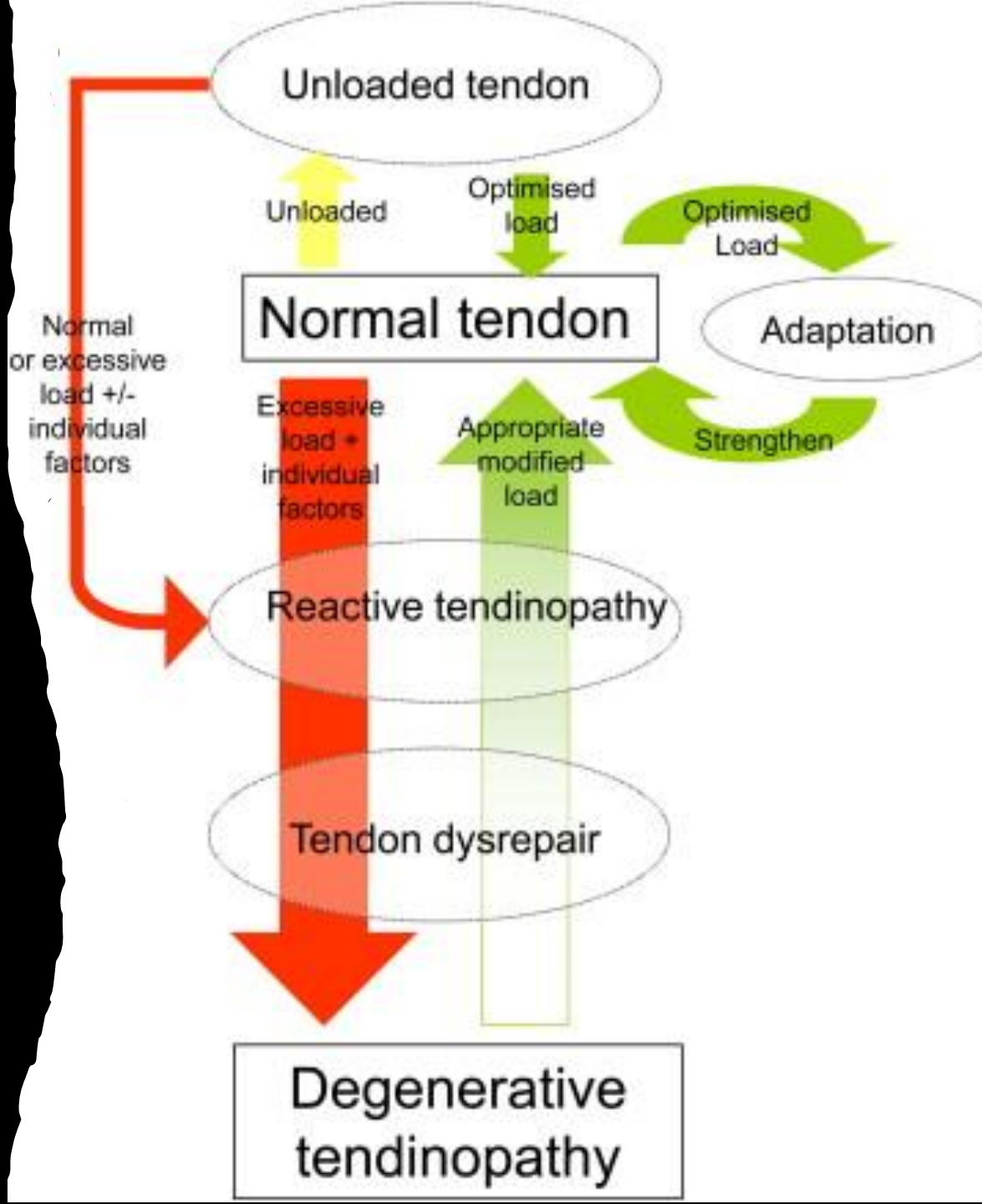
Healing Process- Tendon

- Blood supply, nutrition & mechanical load, can influence the healing of tendon injuries.
- Thomopoulos et al. (2015) emphasized the role of vascularization in tendon healing and the potential effects of aging on blood flow to tendons.

Healing Process- Tendon

- Lack of vascularity:
  - Can cause problems in healing
  - Fibrous union required to provide extensibility and flexibility
    - Abundance of collagen needed to achieve good tensile strength
    - Collagen synthesis can become excessive can result in fibrosis: adhesions from in surrounding structures
      - Interfere with gliding and smooth movement
      - Tensile strength not sufficient to permit strong pull for 4 to 5 weeks
        - At risk of strong contraction pulling tendons ends apart

Patterson-Kane, Janet & Becker, David. (2012). The Pathogenesis of Tendon Microdamage in Athletes: The Horse as a Natural Model for Basic Cellular Research. Journal of comparative pathology. 147. 227-47. 10.1016/j.jcpa.2012.05.010.



Rehabilitation:

- A study by Jonsson et al. (2008) demonstrated the effectiveness of eccentric exercise in the treatment of Achilles tendinopathy.
- A systematic review by Malliaras et al. (2013) found eccentric exercise to be effective in reducing pain and improving function in patients with patellar tendinopathy.
- Eccentric exercises involve lengthening the muscle-tendon unit under load, which can stimulate collagen synthesis and improve tendon strength.

Rehabilitation:

Return to Activity Guidelines

- Gradual return to activity is necessary to prevent re-injury and ensure safe rehabilitation of the tendon.
- A study by Silbernagel et al. (2017) emphasized the importance of:
  - a progressive exercise program
  - functional testing
  - patient education in facilitating a safe and successful return to sport
- Clear guidelines for activity modification, workload management, and monitoring of symptoms help individuals transition back to their desired level of activity.

Rehabilitation:

- Preventing tendon injuries requires addressing risk factors, maintaining proper biomechanics, and implementing injury prevention strategies.
- Coping with a tendon injury and the associated physical limitations can have psychological implications for individuals undergoing rehabilitation.
- Educating patients on the recovery process, setting realistic goals, and providing emotional support can aid in managing the psychological impact of tendon injuries.
- A study by Ardern et al. (2016) emphasized the importance of addressing psychosocial factors in rehabilitation to optimize outcomes and enhance patient well-being.

Rehabilitation:

Pain Management Strategies for Tendon Injuries

- Effective pain management is essential in tendon rehabilitation to improve patient comfort, compliance with treatment, and functional recovery.

- Patient education on pain management techniques, pacing activities, and self-care practices empowers individuals to take an active role in their recovery and well-being.

Outcome Measures

Recovery from tendinopathy can take 1 year
+

Outcome measures include:

Patient-reported pain & symptoms

Muscle performance

Function

Rehabilitation

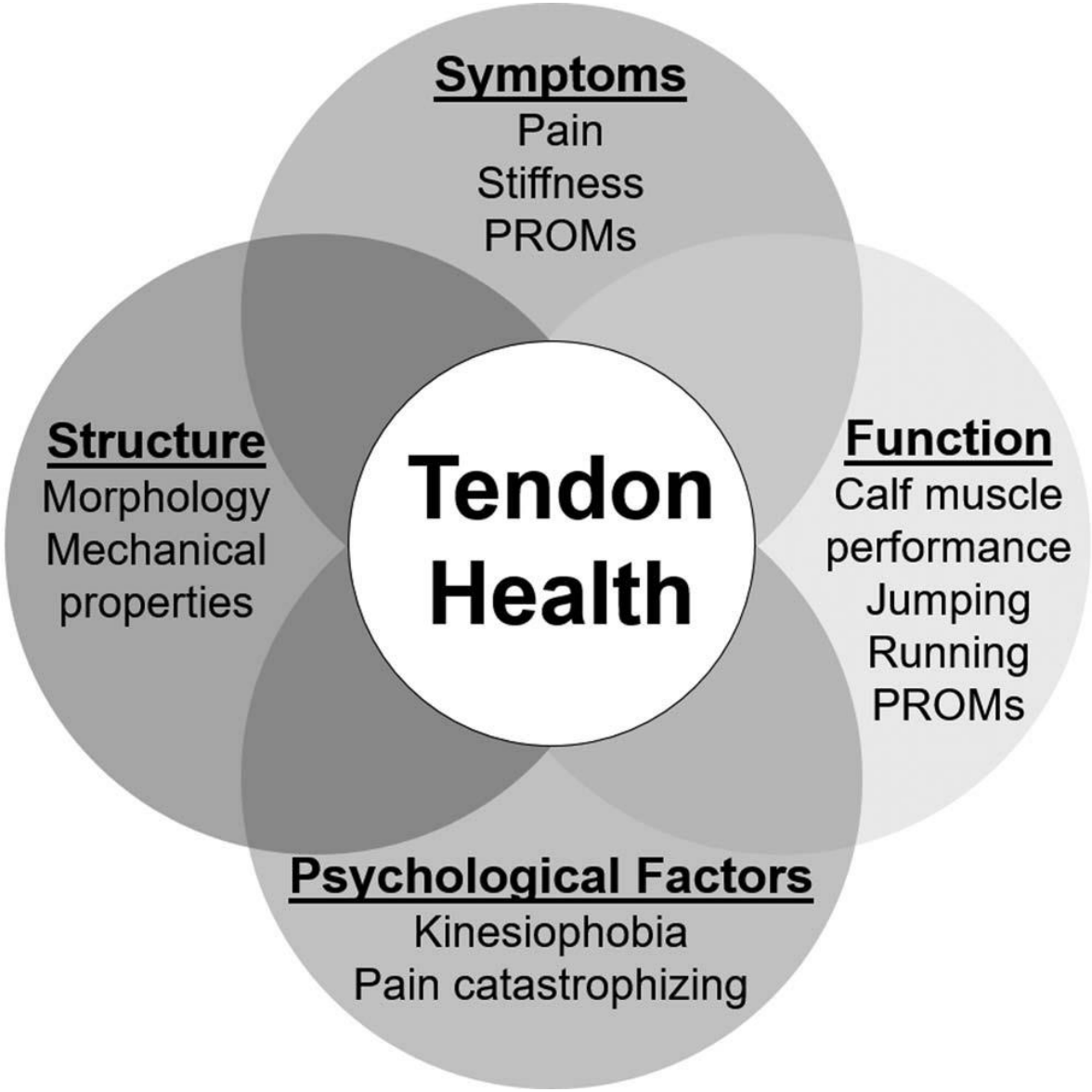
Exercise rehabilitation has the highest level of evidence

The purpose of exercise is to provide mechanical load to the tendon to promote remodelling decrease pain and improve endurance and strength of low leg function

Although more evidence is required, isometric exercises as the initial treatment may be favourable, followed by eccentric (or isolated concentric and eccentric exercises).

Exceeding limits is likely to be detrimental

Domains of Tendon Health



Achilles Tendon Loading Protocol

Symptom-management and load-reduction phase: wk 1 to 2 (or longer if needed)

- Patient status
  - Pain and difficulty with all activities, difficulty performing 10 one-legged heel rises
- Loading intensity
  - Progress loading up to 100% body weight with slow, controlled motion. If needed, begin with aquatic therapy, body-weight support, or isometric plantar flexion.
- Goals
  - Start to exercise and understand nature of the injury and how to use the pain-monitoring model
- Treatment program:
  - Perform exercise once a day
  - Pain-monitoring model information and advice on exercise activity
  - Circulation exercise (moving foot up or down)
  - Two-legged heel rises standing on the floor (3 x 10 to 15 repetitions)
  - One-legged heel rises standing on the floor (3 x 10 repetitions)
  - Eccentric heel rises standing on the floor (3 x 10 repetitions)
  - Sitting heel rises (3 x 1 O repetitions)

Silbernagel, K. G., Hanlon, S., & Sprague, A. (2020). Current clinical concepts: conservative management of Achilles tendinopathy. Journal of athletic training, 55(5), 438-447.

Achilles Tendon Loading Protocol

- **Recovery phase: wk 2 to 5 (or longer if needed)**

- **If pain at the distal insertion of the tendon, continue standing on the floor Patient status**
- **Pain with exercise, morning stiffness, pain when performing heel rises Loading intensity**
  - **The load on the Achilles tendon is increased by increasing the speed of movement and by adding external resistance. External resistance is introduced once the patient can complete the body-weight treatment program without difficulty.**
- **Goals**
    - **Start strengthening**
- **Treatment program: Perform exercises once a day**
    - **Two-legged heel rises standing on edge of a step (3 x 15 repetitions) One-legged heel rises standing on edge of a step (3 x 15 repetitions) Eccentric heel rises standing on edge of a step (3 x 15 repetitions) Sitting heel rises (3 x 15 repetitions)**
    - **Quick-rebounding heel rises (3 x 20 repetitions)**

Silbernagel, K. G., Hanlon, S., & Sprague, A. (2020). Current clinical concepts: conservative management of Achilles tendinopathy. Journal of athletic training, 55(5), 438-447.

Achilles Tendon Loading Protocol

- **Rebuilding phase: wk 3 to 12 (or longer if needed)**

- **If pain at the distal insertion of the tendon, continue standing on the floor**
- **Patient status**
  - **Tolerates the recovery phase exercise program well, no pain at the distal tendon insertion, possibly decreased or increased morning stiffness**
- **Loading intensity**
- **Continue to progress external resistance and initiate plyometric exercises according to patient tolerance. Goals**
    - **Heavier strength training, increase or start running or jumping activity**
- **Treatment program: Perform exercises every day and with heavier load 2 to 3x/wk**
    - **One-legged heel rises standing on edge of step with added weight (3 x 15 repetitions) Eccentric heel rises standing on edge of step with added weight (3 x 15 repetitions) Sitting heel rises (3 x 15 repetitions)**
    - **Quick-rebounding heel rises (3 x 20 repetitions) Plyometrics training (sport specific)**

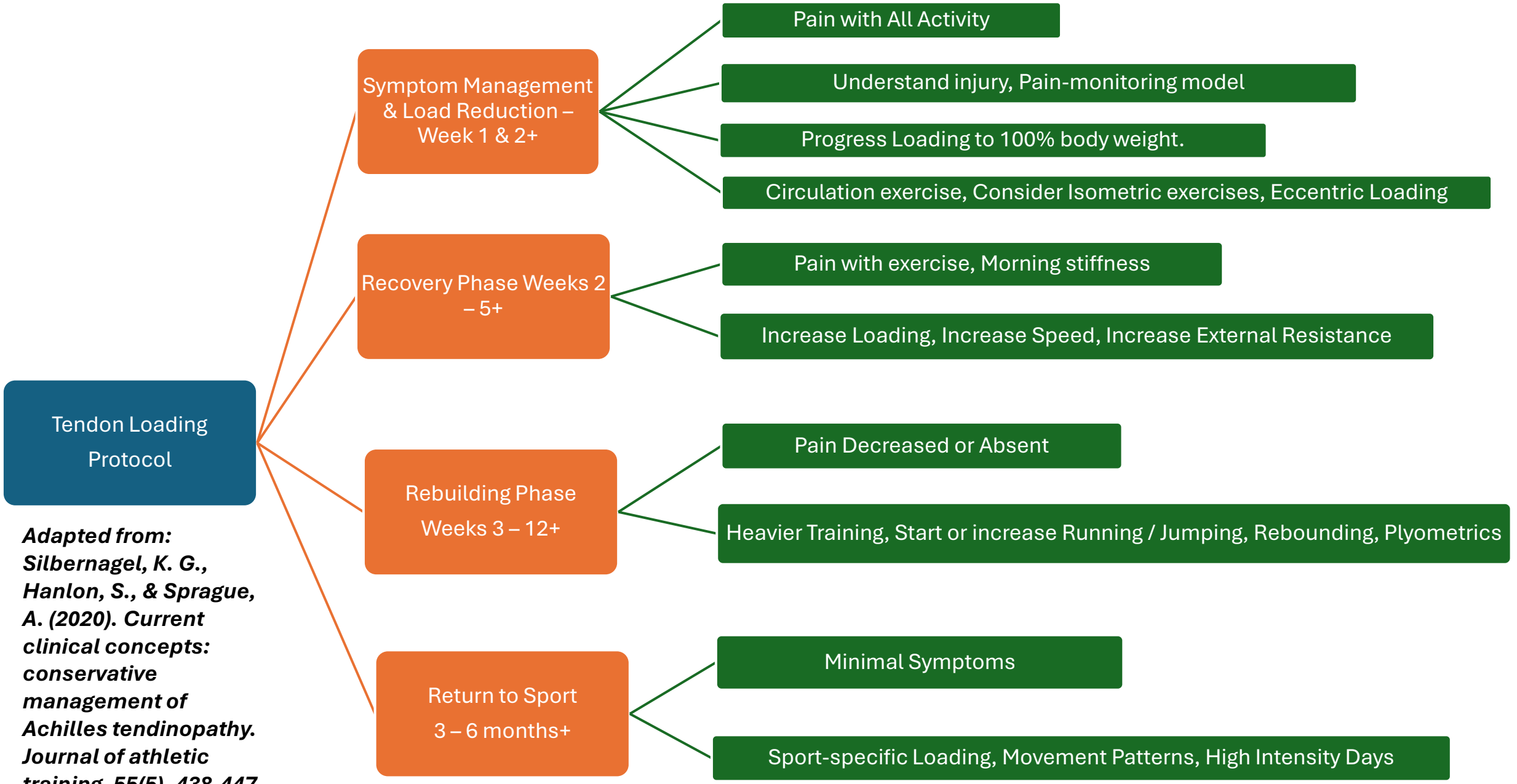
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Achilles Tendon Loading Protocol

- **Return-to-sport phase: 3 to 6 mo (or longer if needed)**

- **If pain at the distal insertion of the tendon, continue standing on the floor Patient status**
- **Minimal symptoms, some but not daily morning stiffness, can participate in sports without difficulty Loading intensity**
- **Progress from previous phase to include sport-specific loading speed and movement patterns on high-intensity days. Goals**
    - **Maintenance exercise, no symptoms**
- **Treatment program: Perform exercises 2 to 3x/wk**
    - **One-legged heel rises standing on edge of step with added weight (3 x 15 repetitions) Eccentric heel rises standing on edge of step with added weight (3 x 15 repetitions) Quick-rebounding heel rises (3 x 20 repetitions)**

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Adapted from: Silbernagel, K. G., Hanlon, S., & Sprague, A. (2020). Current clinical concepts: conservative management of Achilles tendinopathy. Journal of athletic training, 55(5), 438-447.

Anti-inflammatories & Steroid injections?

- In the past, anti-inflammatories and steroid injections were commonly used to alleviate symptoms associated with tendinopathy. Is this effective?
- Anti-inflammatories, such as ibuprofen and naproxen, were traditionally believed to reduce inflammation and pain associated with tendinopathy. In fact, chronic use of anti-inflammatories may actually hinder the natural healing process of the tendon by reducing collagen synthesis and weakening the tendon structure.
- Research has shown that steroids may weaken the tendon structure and increase the risk of tendon rupture. Additionally, steroids may only provide temporary relief and do not address the underlying cause of tendinopathy.
- While anti-inflammatories and steroid injections may provide short-term relief for symptoms of tendinopathy, they are not recommended as long-term treatments. Physiotherapy, eccentric exercise, and other conservative treatments have been shown to be more effective in promoting tendon healing and preventing further injury.

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- Cook, J.L. & Purdam, C.R. (2009). Is tendon pathology a continuum? A pathology model to explain the clinical presentation of load-induced tendinopathy. British Journal of Sports Medicine, 43: 409-416.
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Tarahumara Running Tribe: https://www.youtube.com/watch?v=lQtc0NUvPbk



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